FORMULA 1 / ENGINEERING REPORT

Vehicle Performance Analysis

Lap timing, telemetry and model performance



SILVERSTONE CASE STUDY / 2024 TO 2025 RACE SAMPLE

1 Introduction

A qualifying gap of a tenth of a second tells me who was quicker, but it gives no indication of where the gap opened up. I started with the 2025 British Grand Prix at Silverstone because I could compare the sector times with the speed, throttle and brake traces for the two quickest drivers. I then looked at how pace changed during the race and used six race events to test whether information available before a lap could predict its time.

I treat the models as answers to different questions. The race predictor estimates the next eligible lap time. The physics model follows measured speed over short windows. Principal component analysis (PCA) compresses a speed trace that has already been recorded. The surrogate estimates the output of the simplified simulator on synthetic scenarios. Each task has its own reference and units.

1.1 Data and preparation

I used public OpenF1 timing, stints, weather, pit and race-control records, together with selected car and location telemetry [1]. The saved snapshot was retrieved on 4 November 2026 and contains 7,129 lap records. I used a fixed seed of 42 for the model experiments [2].

Table 1. Session coverage and lap eligibility

Year	Circuit	Session	Source laps	Eligible laps
2024	Catalunya	Race	1,310	1,221
2024	Silverstone	Race	962	869
2024	Spielberg	Race	1,407	1,214
2025	Catalunya	Race	1,204	990
2025	Silverstone	Qualifying	291	169
2025	Silverstone	Race	827	385
2025	Spielberg	Race	1,128	960

I kept the original lap rows and marked their eligibility. To enter the pace analysis, a lap needed valid timing and sector times that reconciled within 0.15 s. I excluded pit and out laps, deleted laps and laps that overlapped a flagged interruption. I treated yellow and red flag messages as 90-second caution intervals and used race-control messages to track safety-car periods. This can exclude a driver who was unaffected by a local flag. The qualifying comparison applies a further 107% pace screen, leaving 88 laps across 20 drivers.

For weather I used the most recent reading at or before lap start, within 120 seconds in the same session. Tyre age counts laps already completed on the set at lap start. Keeping both variables on that timeline mattered later when I built the race predictor.

2 Qualifying pace at Silverstone

Max Verstappen (VER) set the quickest eligible lap at **84.892 s**. Oscar Piastri (PIA) was 0.103 s behind and Lando Norris (NOR) was 0.118 s behind. I used the sector times from each driver's quickest eligible lap so that the sector differences add up to the lap gap.

Piastri lost 0.122 s in Sector 1 and 0.140 s in Sector 2 relative to Verstappen, then recovered 0.159 s in Sector 3. Norris also recovered time in Sector 3. The sector split therefore shows where the two McLarens got closer, even though neither matched Verstappen's overall time.

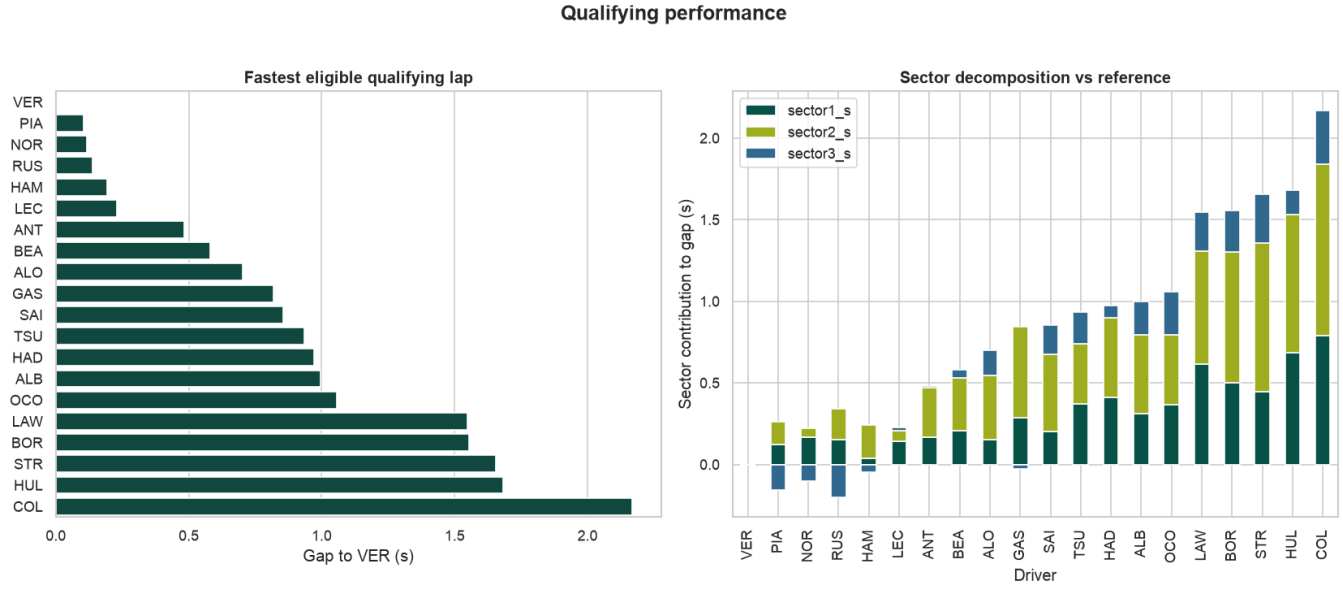


Figure 1. Gap to Verstappen and the sector contributions from each driver's fastest eligible lap.

The lap-time boxes show the spread across the 88 laps that passed the qualifying screen. The driver labels give the sample sizes. Drivers ran in different qualifying phases and had different numbers of attempts, so I read the boxes as a description of the sampled laps.

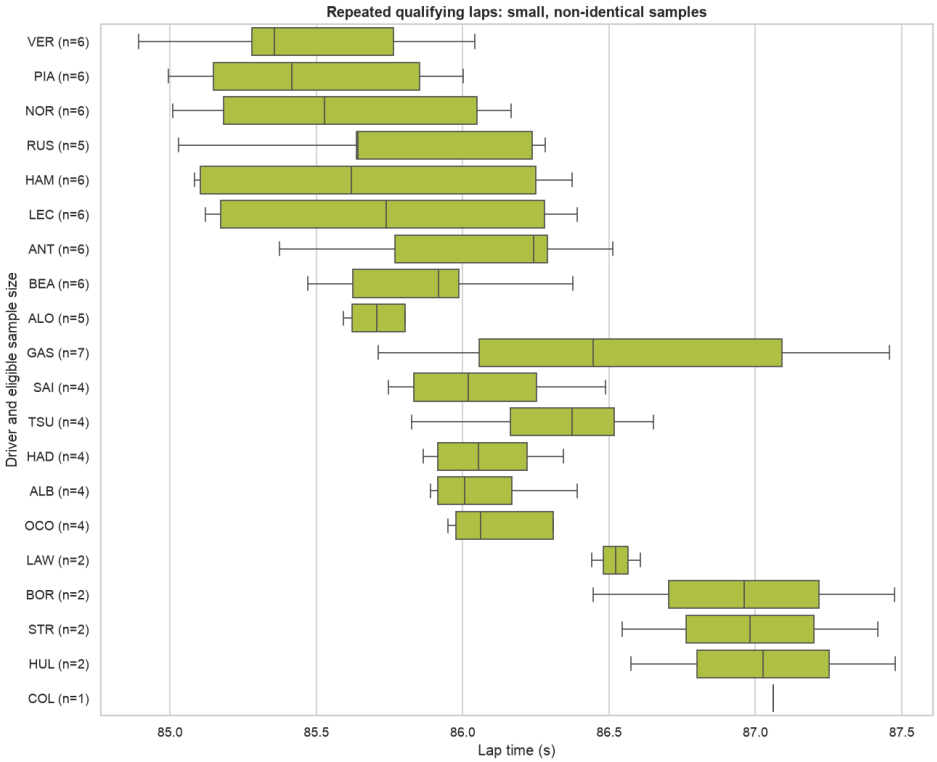


Figure 2. Distribution of screened qualifying laps, with the number of laps shown next to each driver.

3 Telemetry across the lap

I aligned Verstappen's and Piastri's quickest laps by integrating measured speed over time and interpolating onto a 10 m grid. Brake and gear use the last observed value. The track views match location readings to car samples within 0.5 s and use approximate coordinates. The brake trace is an on/off channel displayed on a 0 to 100 scale [1].

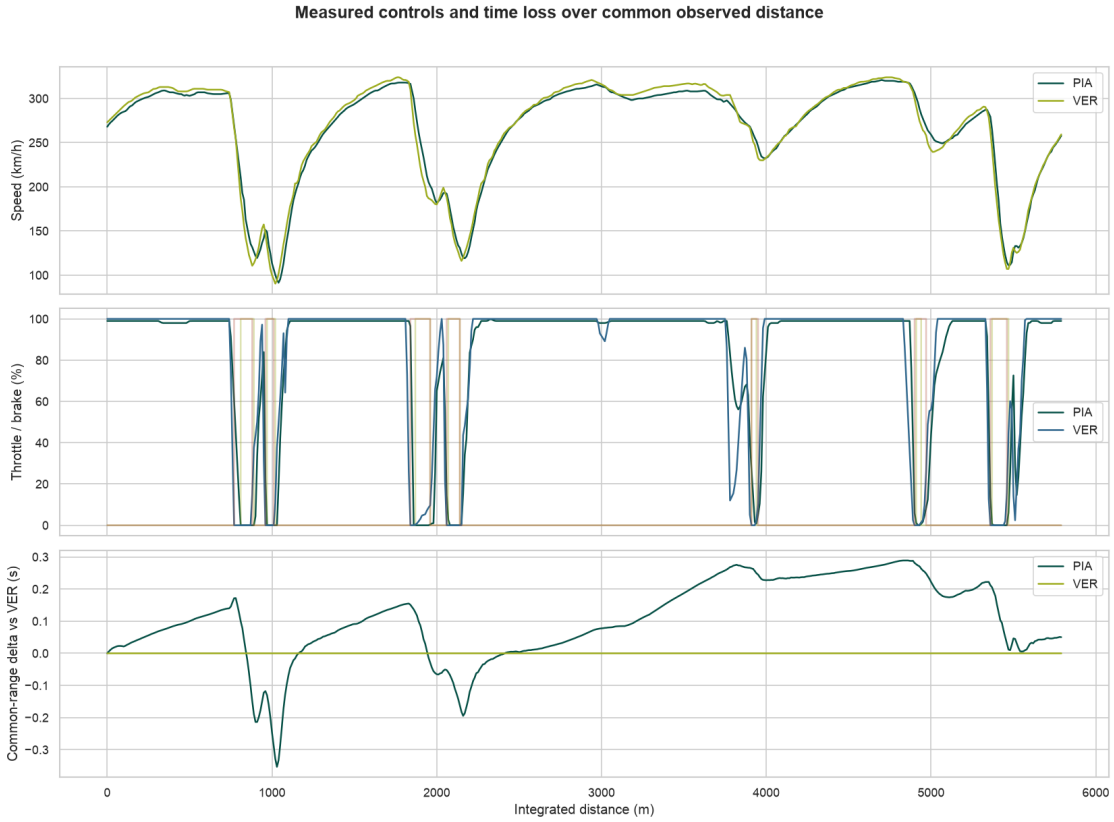


Figure 3. Speed, throttle, brake and cumulative elapsed-time difference across the shared observed distance. A positive delta means Piastri took longer to reach that distance.

The observed-range delta ends at **+0.050 s** for Piastri, compared with the timing gap of **+0.103 s**. I kept the 0.053 s discrepancy because each lap's distance was integrated separately and the samples do not fall exactly on the timing lines. Verstappen's speed trap read 323 km/h against Piastri's 319 km/h, while their time-weighted mean speeds over the observed samples were much closer at 246.838 and 246.732 km/h, respectively.

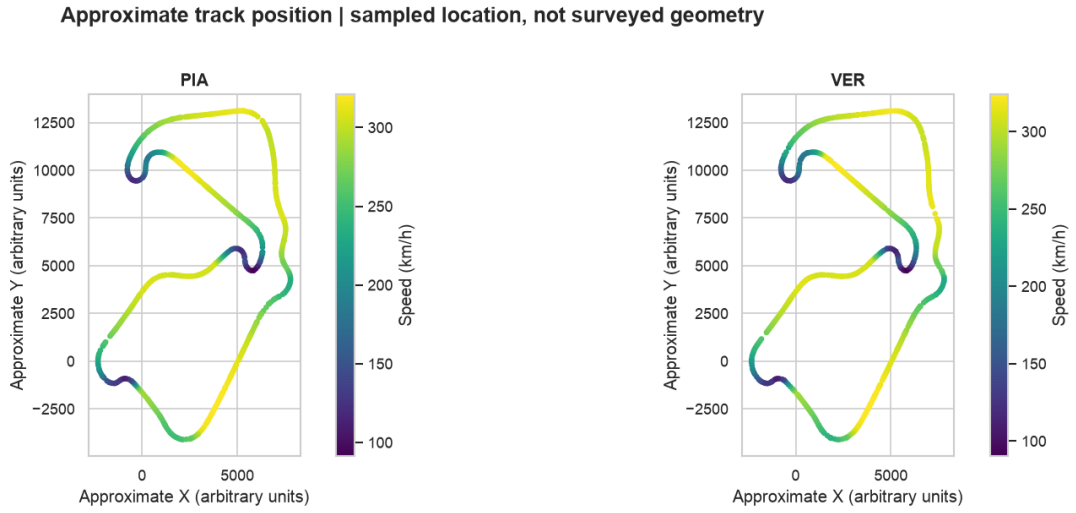


Figure 4. Speed on the two fastest laps, shown against approximate circuit coordinates.

4 Tyres, weather and race pace

The 2025 British race had 385 eligible laps. The intermediate tyre accounted for 264 of them and had a median time of 104.005 s. Mediums accounted for 87 laps with a median of 91.273 s. Hard and soft tyres had smaller samples of 11 and 23 laps, respectively. The compound groups include laps from different stages of the race, so the difference in medians also reflects the conditions in which each tyre was used.

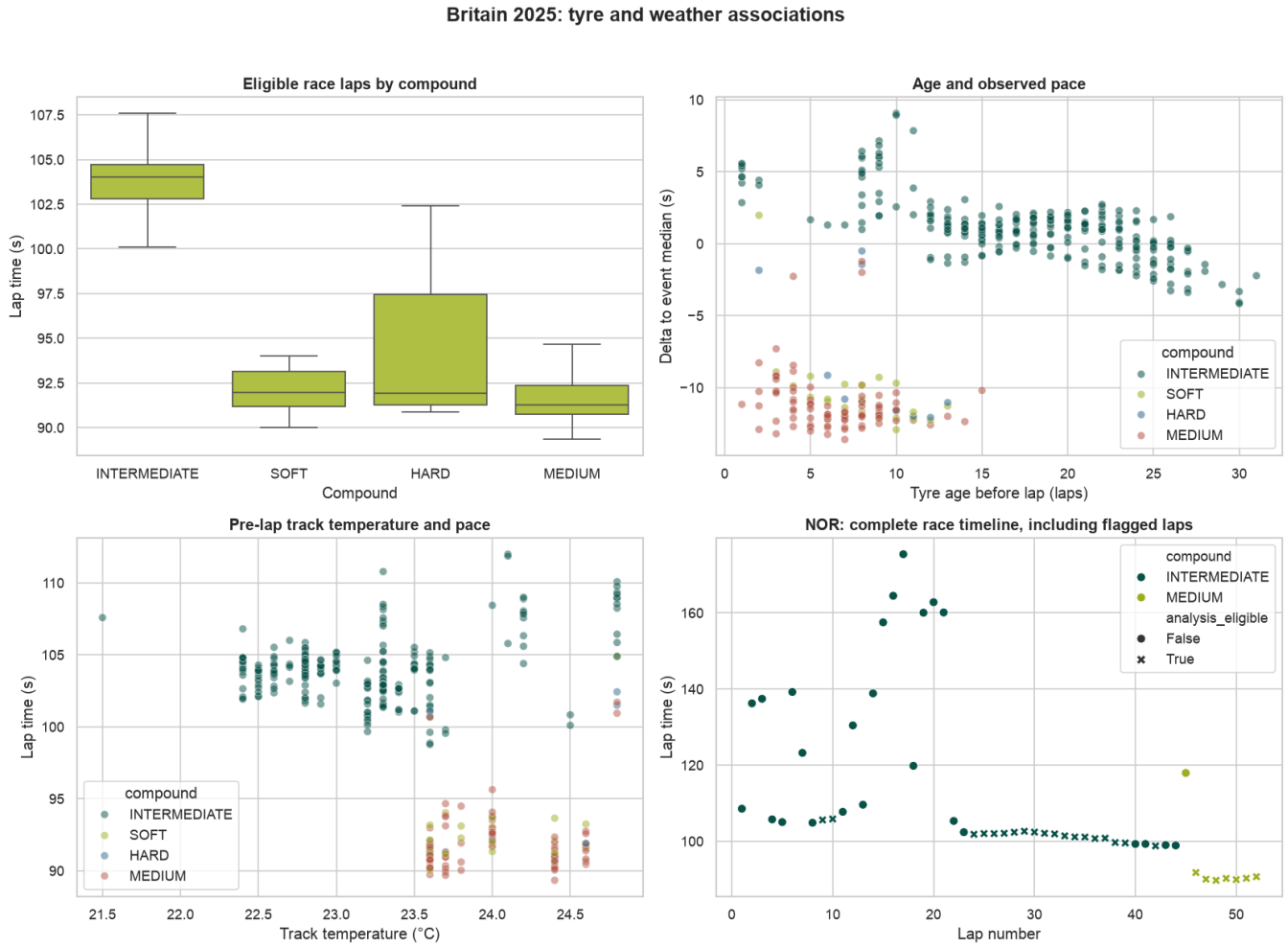


Figure 5. Race pace by compound, tyre age and pre-lap track temperature, followed by Norris's race timeline with flagged laps retained.

I subtracted the eligible race median for the tyre-age plot. That gives the pace on a common scale within the event, while traffic, fuel load and the state of the track can still vary with tyre age. In Norris's timeline, the later medium-tyre laps are quicker than his intermediate-tyre laps. Looking at the sequence alongside the compound boxes makes it easier to see why those groups are a poor basis for estimating a tyre effect on its own.

5 Predicting the next race lap

The prediction is made at lap start. The inputs are circuit, driver, team, race lap number, compound, tyre age, pre-lap weather and the median of the driver's previous three eligible laps. I shifted the median by one lap and excluded the first three eligible laps per driver and session. Earlier eligible laps from a test race can inform its rolling median as the race unfolds. The fitted models stay frozen throughout the race.

I trained on the 2024 races at Catalunya, Silverstone and Spielberg (3,127 laps). I used Catalunya 2025 for validation (933 laps), then tested on Silverstone and Spielberg 2025 (1,245 laps). I compared a training circuit median, the prior three-lap median, ridge regression and a random forest. Ridge and random forest estimate the adjustment to the prior median. The final scores cover eligible completed laps.

Table 2. Race lap-time prediction errors in seconds

Model	Val. MAE	Val. RMSE	Test MAE	Test RMSE	Test max.
Training circuit median	1.124	1.443	2.625	4.601	17.886
Prior 3-lap median	0.570	0.894	**0.728**	1.610	10.524
Ridge, alpha = 1	1.285	1.486	0.891	**1.356**	**9.029**
Random forest, leaf = 5	**0.497**	**0.759**	1.048	1.971	10.375

MAE is the mean absolute error. RMSE puts more weight on large misses, and the maximum column gives the largest absolute error. The forest led on validation. On the two test races, the prior three-lap median had the lowest MAE, while ridge had the lowest RMSE and maximum error. The choice depends on how much weight I give to the larger misses.

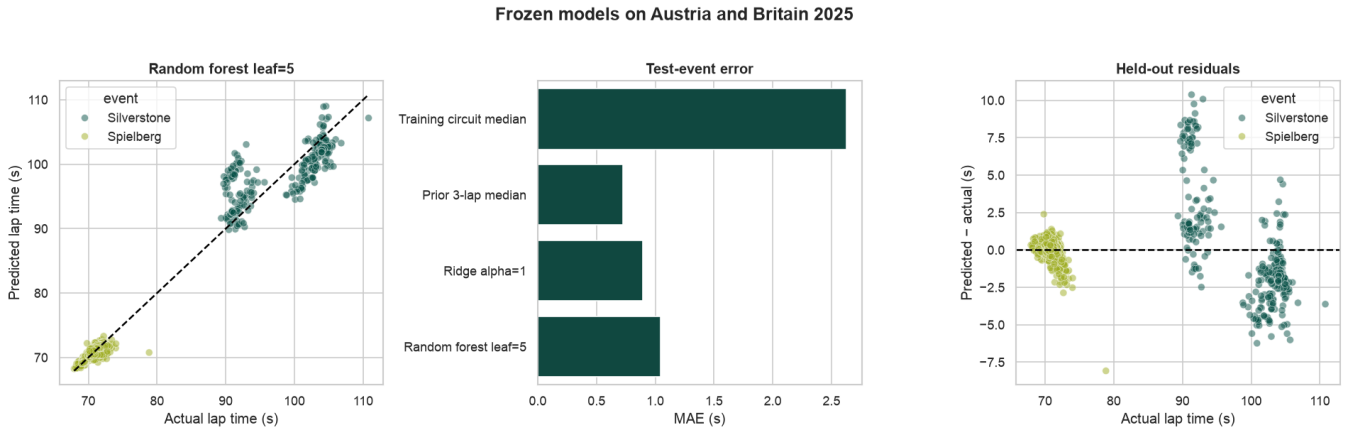


Figure 6. Actual and predicted lap times for the selected forest, test MAE for the four models and signed residuals by circuit.

5.1 Where the forest struggled

The forest's MAE was **0.359 s at Spielberg** and **2.912 s at Silverstone**. Spielberg accounts for 909 of the 1,245 test laps, so the pooled result is weighted towards that event. Silverstone's residuals are more spread out, with errors on both sides of zero. The changing race conditions provide context for the difference, although two test races give me little basis for separating the possible causes.

Table 3. Held-out random forest results by race

Race	Laps	MAE (s)	RMSE (s)	Max. abs. (s)
Silverstone 2025	336	2.912	3.673	10.375
Spielberg 2025	909	0.359	0.580	8.081

I also shuffled each input on the Spain validation set to check what the forest relied on. Shuffling tyre age raised MAE by 0.135 s, followed by the prior-lap median at 0.057 s and race lap number at 0.042 s. These values describe dependence within this fitted model and this validation race. The prior median remains the baseline when it is shuffled as an input to the forest's adjustment.

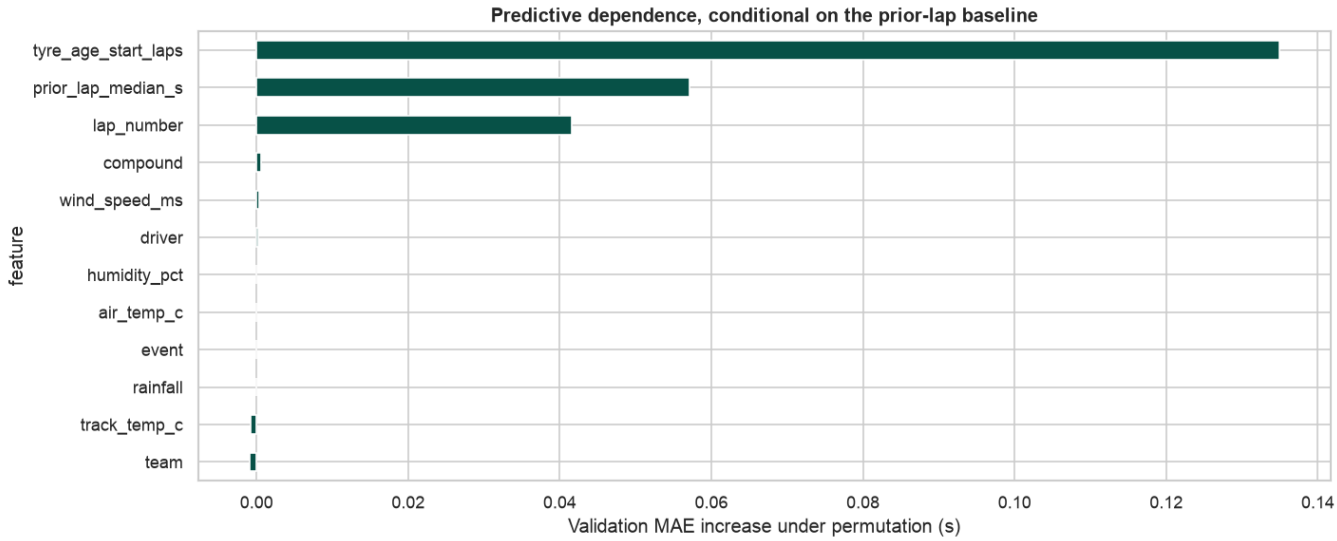


Figure 7. Permutation importance for the forest's adjustment on the Spain validation race. The bars show the change in MAE when each input is shuffled.

6 A short-window model of speed

I used a two-state model for speed and distance, with throttle as a fraction and the brake switch as a 0 or 1 input.

$$\dot{v} = a_T u_T - a_B u_B - k_D v^2 - a_R, \qquad \dot{s} = v$$

The coefficients represent effective acceleration under throttle, deceleration when braking, a speed-squared term and a constant loss. I fitted them by minimising the error in integrated speed. Each five-second window starts at the measured speed. Windows with sample gaps above 0.75 s were rejected, and train, validation and test windows came from disjoint laps. Validation selected the robust-loss scale of 3 m/s.

Table 4. Speed errors on the qualifying windows

Split	Model	MAE (m/s)	RMSE (m/s)	Max. abs. (m/s)	Samples
Validation	Constant speed	11.128	16.316	50.556	528
Validation	Longitudinal physics	2.101	3.224	16.095	528
Test	Constant speed	12.423	17.516	51.111	505
Test	Longitudinal physics	**2.218**	**3.474**	**18.112**	505

The physics model cut test MAE by about 82% relative to holding the window's initial speed constant. That result covers 28 windows from two qualifying laps, with 505 scored speed samples after dropping the reset points. The samples within a window are serially related. Braking had the highest MAE at 4.814 m/s, compared with 1.370 m/s under high throttle and 2.892 m/s on partial throttle or lift.

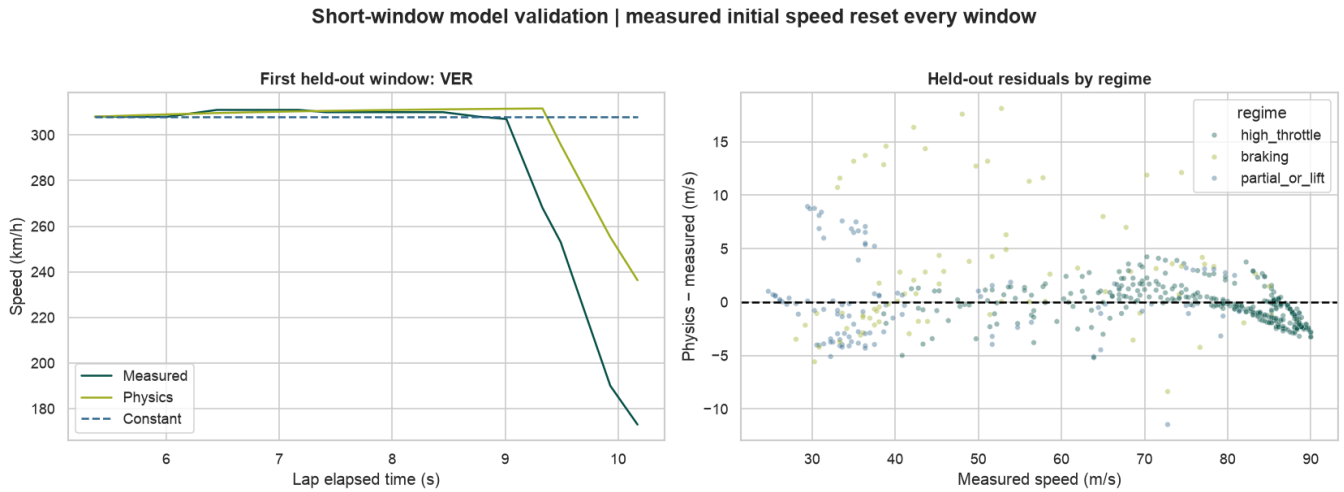


Figure 8. Example held-out speed window and the spread of speed residuals by driving regime.

6.1 Acceleration, gear and model limits

The acceleration plot compares a smoothed derivative of measured speed with the acceleration implied by the model. The gear trace provides context for abrupt changes. Breaks in the plot separate windows, since each one starts from a measured speed. The fitted model captures the broad accelerations and decelerations, but it misses several sharp braking peaks.

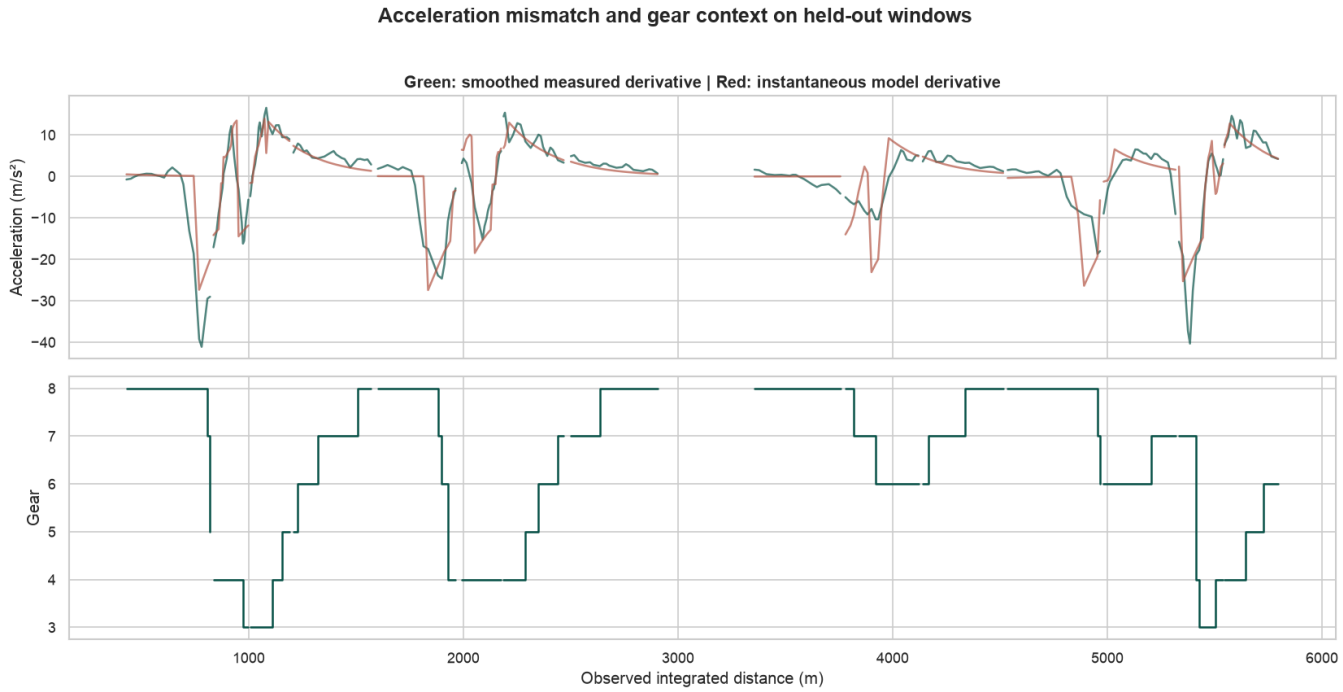


Figure 9. Observed and modelled acceleration with gear on Verstappen's held-out windows. The observed acceleration is derived from speed with a centred smoothing filter.

The fitted throttle and braking coefficients were 16.957 and 10.481 m/s^2. The speed-squared coefficient was 0.002250 per metre, and the constant loss reached its lower bound of zero. The fit's Jacobian condition number was about 74,476. I treat the coefficients as a useful fit to the available traces. Brake pressure, steering, slope and gear-dependent power are outside this model.

On validation windows, changing one coefficient at a time made the fit most sensitive to the throttle and speed-squared terms. Increasing the throttle coefficient by 10% raised MAE from 2.101 to 3.162 m/s. That is a local sensitivity check around the fitted values. The frequent resets and the limited lap sample leave full-lap simulation accuracy open.

7 Compressing the observed speed profiles

I fitted PCA on distance-aligned training laps, then compared two, three and five components on separate validation laps. Five components had the lowest validation RMSE at 3.649 km/h and accounted for 99.51% of the variation in the training profiles. Across the two held-out laps, reconstruction MAE was 2.098 km/h and RMSE was 2.953 km/h.

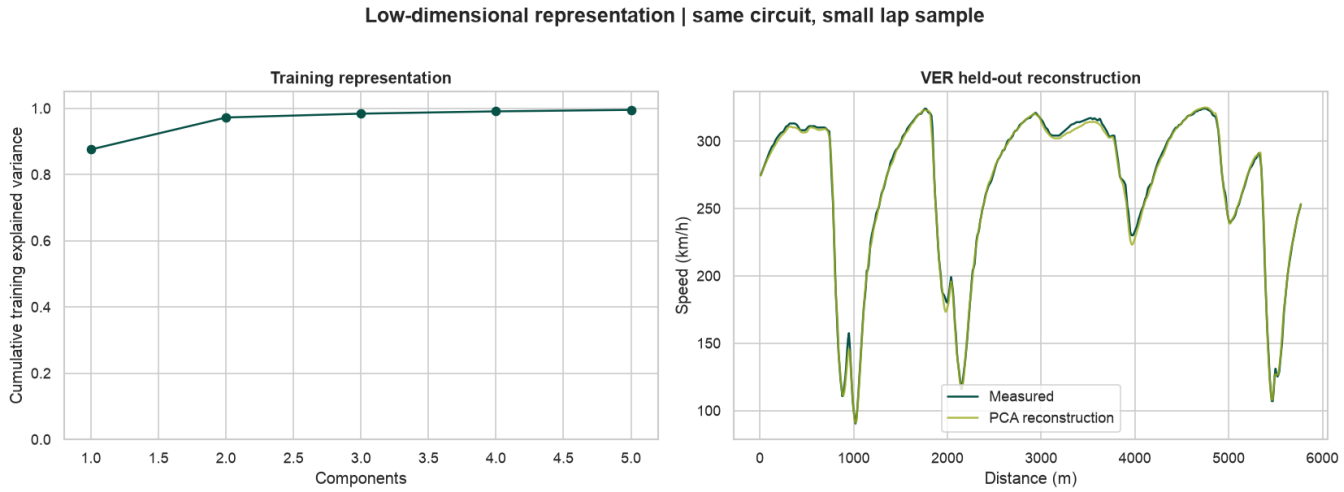


Figure 10. Cumulative explained variance on the training laps and an example reconstruction on a held-out lap.

The reconstruction uses the observed speed trace to calculate its component scores. Its role here is to describe a recorded lap with fewer values. I would need a separate test across circuits and conditions to know whether the same representation transfers.

8 Approximating the simulator

I generated 2,400 scenarios with constant throttle or brake inputs, durations of one to six seconds and starting speeds from 20 to 90 m/s. The target is final speed from the Euler simulator with a maximum time step of 0.05 s. I trained below 60 m/s, used 60 to 70 m/s for validation and reserved 70 to 90 m/s for testing. A degree-three polynomial ridge model had the lowest validation MAE.

Table 5. Errors for speed-profile reconstruction and simulator emulation

Task and split	Model	Unit	MAE	RMSE	Max. abs.	n
PCA test	Five components	km/h	2.098	2.953	18.584	1,154
Surrogate validation	Degree-three ridge	m/s	0.988	1.356	5.921	304
Surrogate test	Degree-three ridge	m/s	2.750	3.636	15.525	701

The PCA sample count is the number of distance bins across two held-out laps. The surrogate sample count is the number of synthetic scenarios. Each error is expressed in the units of its own target.

8.1 Testing the surrogate at higher speeds

At 70 to 90 m/s, the surrogate's error was largest under braking. The braking scenarios had a mean absolute error of 4.370 m/s, against 1.721 m/s for lift and 2.316 m/s for powered scenarios. The plot shows how the signed errors change across the starting-speed range.

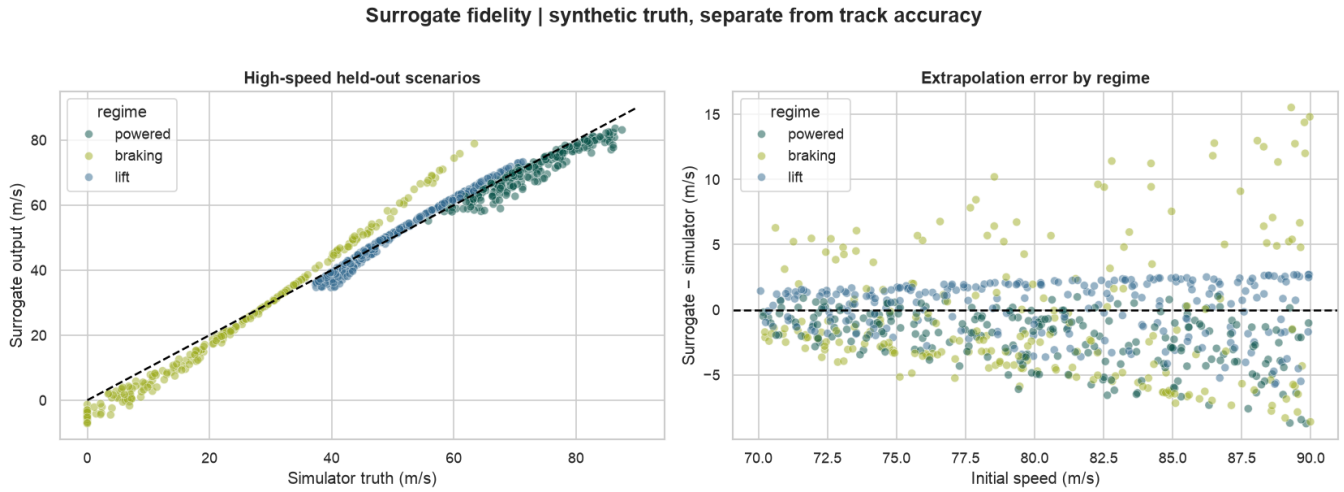


Figure 11. Predicted versus simulated final speed and signed errors for the high-speed synthetic test set.

On the notebook's machine, the median time for a batch of 128 scenarios was 6.23 ms for Euler integration and 0.42 ms for the surrogate over five warmed-up runs. Those timings describe this batch and implementation. The higher-speed errors matter when deciding whether the speed-up is worth using.

9 Conclusion

The Silverstone analysis showed where Piastri and Norris recovered time in the final sector and where Piastri gained and lost time along the sampled telemetry trace. The difference between the trace endpoint and the official gap gave me a practical limit on how precisely I could interpret the distance alignment.

For race prediction, the prior three-lap median was the strongest benchmark on test MAE, while ridge performed better on RMSE. The forest's much larger errors at Silverstone show why I would keep the event-level results alongside the pooled score. On the engineering side, the short-window physics model improved on constant speed, while the acceleration traces exposed its braking errors. PCA was effective for compressing recorded laps at Silverstone, and the surrogate was quicker in the measured batch but less accurate at higher starting speeds. Each result is tied to the conditions and reference used to measure it.

9.1 Sources

[1] OpenF1 API documentation. Historical timing, car data, location, race control, stints and weather. `https://openf1.org/docs/`

[2] F1 Vehicle Performance Analysis. Executed Jupyter notebook, 4 November 2026. Source for the figures, experiment definitions and recorded results.